

IIT JAMMU SUMMER SCHOOL 2026

Course Name : IoT and Drone Building

IoT Assignment 3

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Roll no.: IoT001

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Submitted to: Mr. Majid Majeed

Repository Link:

<https://github.com/Subham827-all/-iot-summer-school-2026>

Ans 36:

MQTT vs HTTP Comparison

Feature	MQTT	HTTP
Architecture Pattern	Publish/Subscribe through an MQTT broker	Client/Server (Request–Response)
Data Transfer Model	Asynchronous; devices publish to topics and subscribers receive updates	Synchronous; client sends a request and waits for the server's response
Power Consumption	Very low due to lightweight packets and persistent connections	Higher because each request requires additional protocol overhead
Latency	Low latency; suitable for real-time communication	Higher latency due to repeated request-response cycles
Use Case Suitability for IoT	Excellent for IoT applications such as smart homes, industrial monitoring, agriculture, and sensor networks	Better suited for web services, REST APIs, dashboards, and occasional data retrieval
Security Considerations	Supports username/password authentication, TLS/SSL encryption, and access control through brokers	Supports HTTPS (TLS/SSL), authentication tokens, API keys, OAuth, and other web security mechanisms

For a **1000-node smart agriculture deployment**, **MQTT** is the preferred communication protocol.

Reasons:

1. **Low Bandwidth Usage:** MQTT uses a lightweight messaging format, reducing network traffic even with thousands of sensor nodes.

2. **Low Power Consumption:** Agricultural sensors are often battery- or solar-powered. MQTT minimizes communication overhead, helping devices operate longer between charges.
3. **Low Latency:** Sensor readings such as soil moisture, temperature, and humidity can be delivered quickly to the central monitoring system.
4. **Scalability:** A single MQTT broker can efficiently manage communication with thousands of connected devices using topics.
5. **Reliable Communication:** MQTT supports different Quality of Service (QoS) levels (0, 1, and 2), allowing developers to choose the appropriate balance between reliability and network overhead.
6. **Efficient Data Distribution:** Multiple applications (e.g., irrigation control, weather dashboard, and mobile app) can subscribe to the same sensor topics without increasing the workload on sensor nodes.
7. **Reduced Network Traffic:** Devices only publish data when necessary, and subscribers receive updates automatically, eliminating the need for continuous polling as required by HTTP.

Ans 37:

Wi-Fi Security for IoT

Wi-Fi-enabled IoT devices such as the ESP32 are vulnerable to several wireless attacks if not properly secured. One common attack vector is exploiting weaknesses in **WPA2**, particularly through weak passwords or misconfigured networks. Attackers may also use an **evil twin attack**, where they create a fake Wi-Fi access point with the same network name (SSID) as a trusted network. If an IoT device connects to the fake access point, attackers can intercept data or steal login credentials.

To improve the security of ESP32-based IoT systems, developers should follow several best practices. Using **WPA3** instead of WPA2 provides stronger encryption and protection against password guessing attacks. When communicating with cloud services, **HTTPS with certificate validation or certificate pinning** should be used to verify the server's identity and prevent man-in-the-middle attacks. Regular firmware updates, strong unique passwords, and disabling unused network services further enhance device security.

Hardcoding Wi-Fi passwords or API keys directly into the source code is dangerous because anyone with access to the code or firmware can retrieve these credentials. If the code is uploaded to a public GitHub repository, attackers can easily obtain sensitive information and gain unauthorized access to the network. A safer approach is to store credentials in a separate configuration file (such as config.h) that is excluded from version control using .gitignore, or use secure storage mechanisms provided by the device.

References:

1. Wi-Fi Alliance. *WPA3™ Security*. <https://www.wi-fi.org/discover-wi-fi/security>
2. Open Worldwide Application Security Project (OWASP). *OWASP Internet of Things Project*. <https://owasp.org/www-project-internet-of-things/>

Ans 38:

LoRa & LPWAN – Long Range IoT

LPWAN (Low Power Wide Area Network) is a wireless communication technology designed for IoT devices that need to send small amounts of data over long distances while consuming very little power. It is commonly used in agriculture, smart cities, and environmental monitoring.

LoRa (Long Range) achieves long-range communication by using a **Chirp Spread Spectrum (CSS)** modulation technique. It also uses a **Spreading Factor (SF)**, where a higher SF increases communication range and reliability but lowers data rate. This allows battery-powered devices to transmit data over several kilometers while using minimal energy.

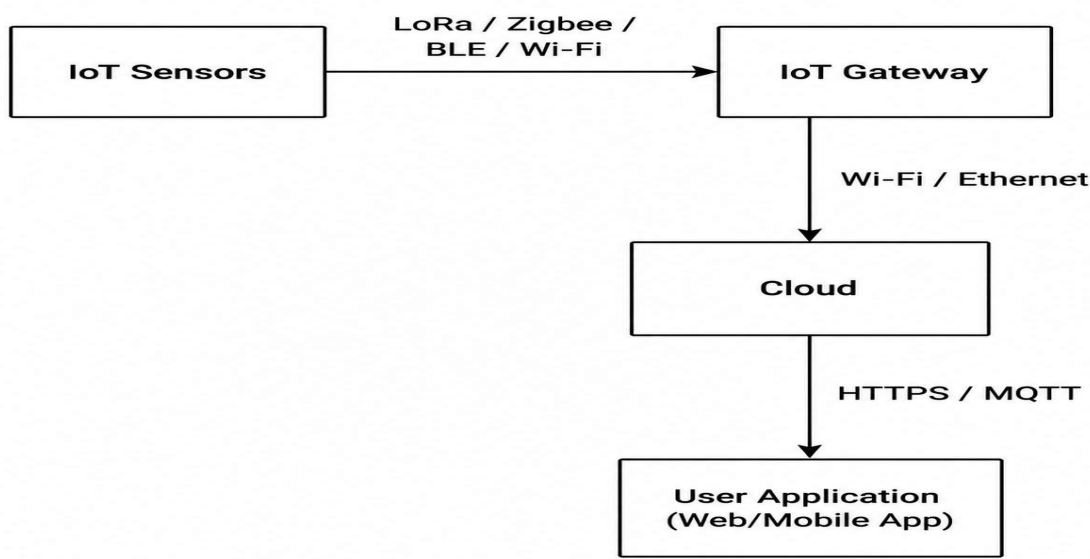
Feature	LoRa	Zigbee	Z-Wave
Range	2–15 km	10–100 m	30–100 m
Power Consumption	Very Low	Low	Low
Data Rate	Low	High	Medium
Best Use	Smart agriculture, smart cities	Home automation	Home automation

A real-world example in India is the **Smart City Mission**, where LoRa-based sensor networks are deployed in cities such as **Pune** and **New Delhi** for applications like smart street lighting, water metering, waste management, and environmental monitoring. LoRa's long range, low power consumption, and ability to connect thousands of devices make it an ideal choice for large-scale IoT deployments.

Ans 39:

IoT Gateway

An **IoT gateway** is a device that connects IoT sensor nodes to the internet or cloud. It collects data from sensors, converts different communication protocols if needed, performs basic processing, and securely forwards the data to cloud services. The gateway also sends commands from the cloud back to the IoT devices.



Protocols Used

Layer	Common Protocols
IoT Sensor → Gateway	LoRa, Zigbee, BLE, Wi-Fi
Gateway → Cloud	MQTT, HTTP/HTTPS, CoAP
Cloud → User Application	HTTPS, MQTT, WebSockets

Ans 40:

MQTT QoS (Quality of Service) Levels

QoS Level	Description	Reliability
QoS 0 – At most once	Message is sent only once without confirmation. It may be lost if the network fails.	Lowest
QoS 1 – At least once	Message is delivered at least once. The receiver acknowledges it, but duplicate messages may occur.	Medium
QoS 2 – Exactly once	Message is delivered exactly once using a four-step handshake. No duplicates occur.	Highest

Which QoS would I choose for an IoT Water Level Monitoring System?

I would choose **QoS 1 (At least once)** because it provides reliable message delivery while keeping communication overhead low. In a water level monitoring system, receiving duplicate readings occasionally is acceptable, but losing critical water level data could lead to incorrect monitoring or delayed alerts. QoS 2 offers the highest reliability but requires more bandwidth and processing, making it less efficient for most IoT applications. Therefore, **QoS 1 provides the best balance between reliability and performance.**

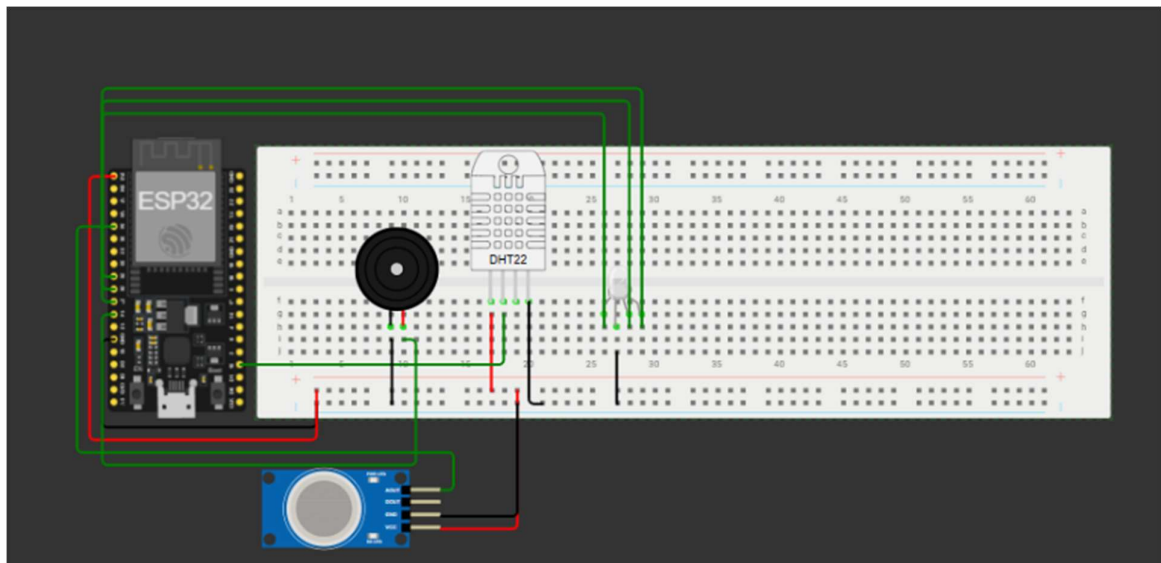
Q41. Smart Home Air Quality Monitor Problem

Statement

A family in a Milan(Europe) home is concerned about indoor air quality during winter when they burn wood for heating. Design an IoT-based Air Quality Monitoring System that continuously measures CO/smoke levels (MQ-2 sensor), temperature and humidity (DHT11), and alerts residents when air quality drops below safe thresholds. The system must work without the internet when the router is down.

GITHUB LINK : <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/QUES41>

CIRCUIT DIAGRAM :



Q42. Automated Plant Watering System

Problem Statement

A J&K horticulture farmer wants to automate irrigation for a small saffron farm plot. The system should read soil moisture every 30 minutes and activate a water pump only when the soil is dry. Saffron is sensitive to overwatering — the system must also prevent watering if it rained recently (use LDR to detect low light/overcast sky as a rain proxy and also leverage the use of api also). The farmer must be able to manually override via a button.

GITHUB LINK: <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/QUES42>

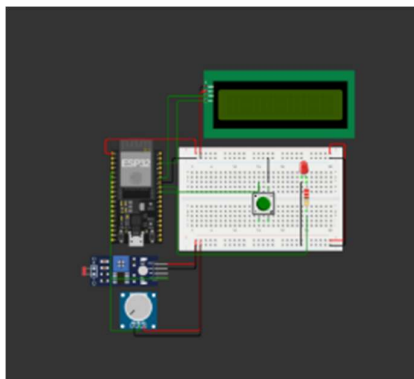


IMAGE :

WOKWI LINK : <https://wokwi.com/projects/469012526400405505>

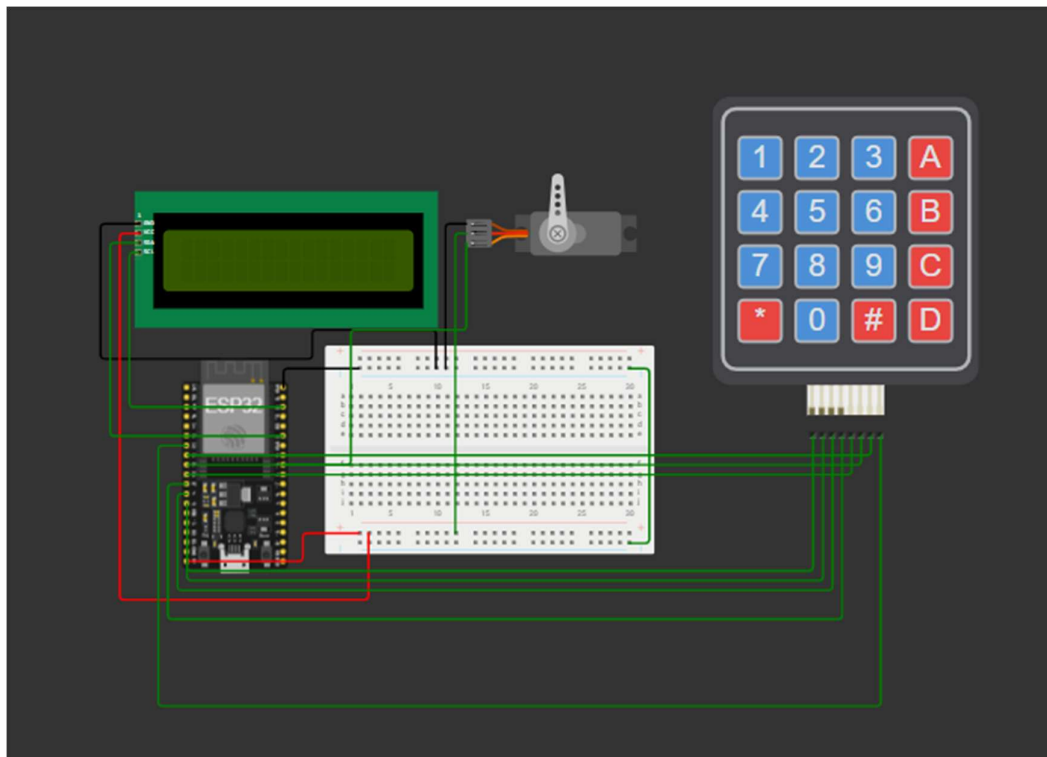
Q43. Smart Door Lock with OTP

Problem Statement

A university hostel wants to replace physical keys with a PIN-based entry system that also generates a One-Time Password. Design a system where a student enters their 4-digit ID on a keypad, the system generates a 6-digit OTP, sends it over Bluetooth to their phone, and only grants access if they enter the OTP within 30 seconds.

GITHUB LINK : <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/QUES43>

CIRCUIT :



WOKWI LINK : <https://wokwi.com/projects/469002185062883329>

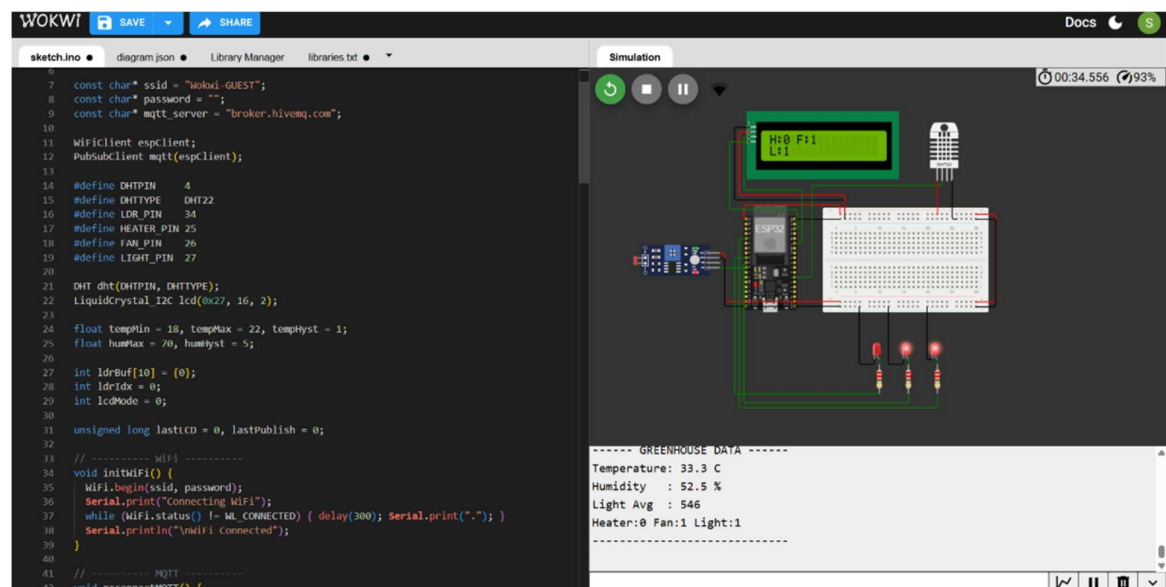
Q45. Smart Greenhouse Controller

Problem Statement

Floriculture farmers in Kashmir growing tulips and roses need automated greenhouse climate control. Design a multi-parameter monitoring and control system that maintains optimal growing conditions: Temperature 18–22°C, Humidity 60–70%, and sufficient light for 12 hours per day.

GITHUB LINK : <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/QUES45>

IMAGE:





Need a fully managed MQTT broker?

Get your own Cloud broker and connect up to 100 devices for free.

[Get your free account](#)

Connection

connected

Publish

Topic

testtopic1

QoS

0

Retain

Publish

Message

Messages

1	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
2	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
3	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
4	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
5	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
6	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
7	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
8	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
9	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
10	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
11	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
12	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
13	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
14	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
15	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
16	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
17	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
18	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
19	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
20	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
21	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
22	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
23	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
24	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
25	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
26	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
27	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
28	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
29	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
30	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
31	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
32	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
33	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
34	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
35	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
36	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
37	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
38	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
39	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
40	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
41	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
42	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
43	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
44	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
45	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
46	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
47	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
48	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
49	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
50	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
51	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
52	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
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54	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
55	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
56	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
57	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
58	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
59	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
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61	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
62	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
63	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
64	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
65	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
66	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
67	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
68	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
69	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
70	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
71	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
72	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
73	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
74	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
75	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
76	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
77	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
78	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
79	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
80	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
81	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
82	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
83	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
84	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
85	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
86	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
87	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
88	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
89	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
90	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
91	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
92	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
93	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
94	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
95	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0
96	2024-07-09 03:18:33	Topic: greenhousehumidity	QoS: 0
97	2024-07-09 03:18:33	Topic: greenhouseheater	QoS: 0
98	2024-07-09 03:18:33	Topic: greenhousefan	QoS: 0
99	2024-07-09 03:18:33	Topic: greenhouselight	QoS: 0
100	2024-07-09 03:18:33	Topic: greenhousetemp	QoS: 0

Subscriptions

Add New Topic Subscription

QoS: 0

greenhouse/temp

X

QoS: 0

greenhouse/humidity

X

QoS: 0

greenhouse/light

X

QoS: 0

greenhouse/heater

X

QoS: 0

greenhouse/fan

X

QoS: 0

greenhouse/lightstatus

X

WOKWI LINK : <https://wokwi.com/projects/469016089565727745>

Q46. River Water Level Alert (Flood Detection)

Problem Statement

Given the frequent flooding near the Tawi and Chenab rivers, design a low-cost IoT flood-early-warning sensor node using an HC-SR04 ultrasonic sensor to monitor water level in a model river (a glass tank). Trigger alerts at predefined levels and transmit data wirelessly.

GITHUB LINK : <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/%20QUES46>

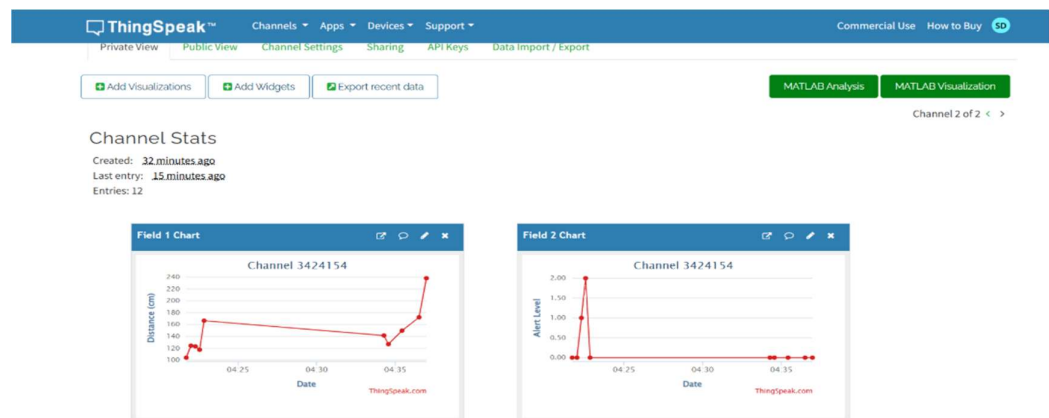
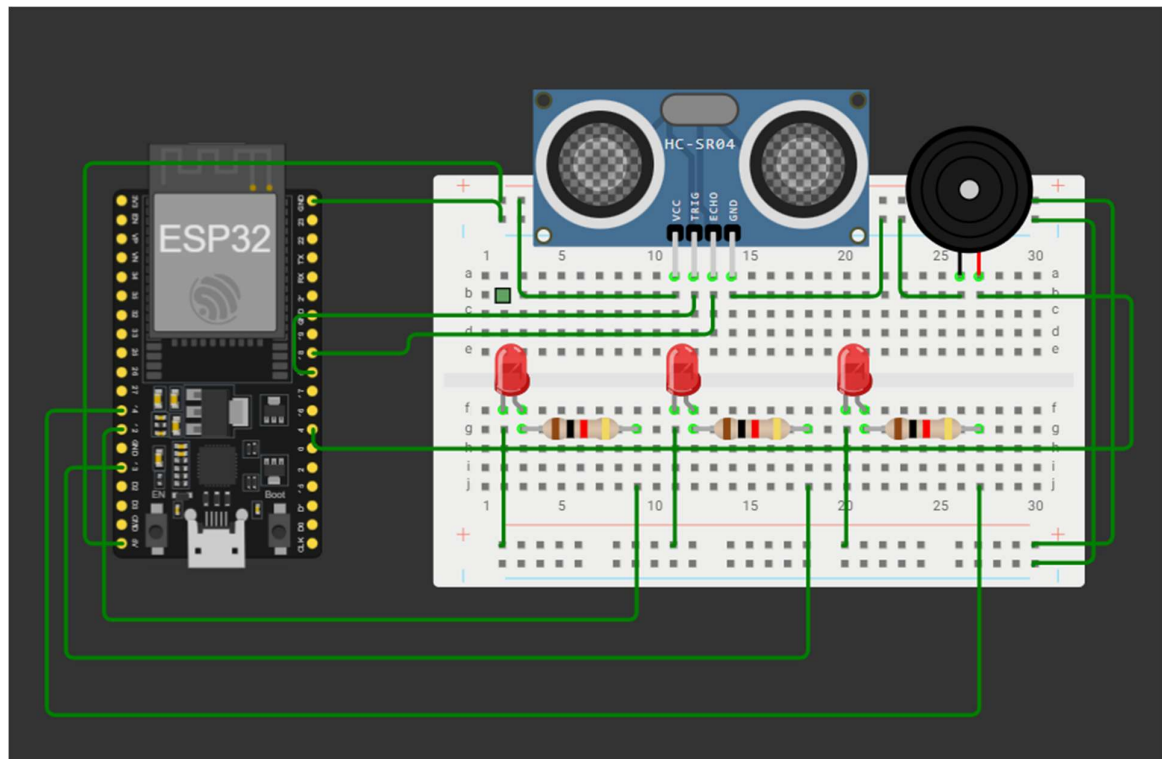


IMAGE :



WOKWI LINK : <https://wokwi.com/projects/469012760047283201>

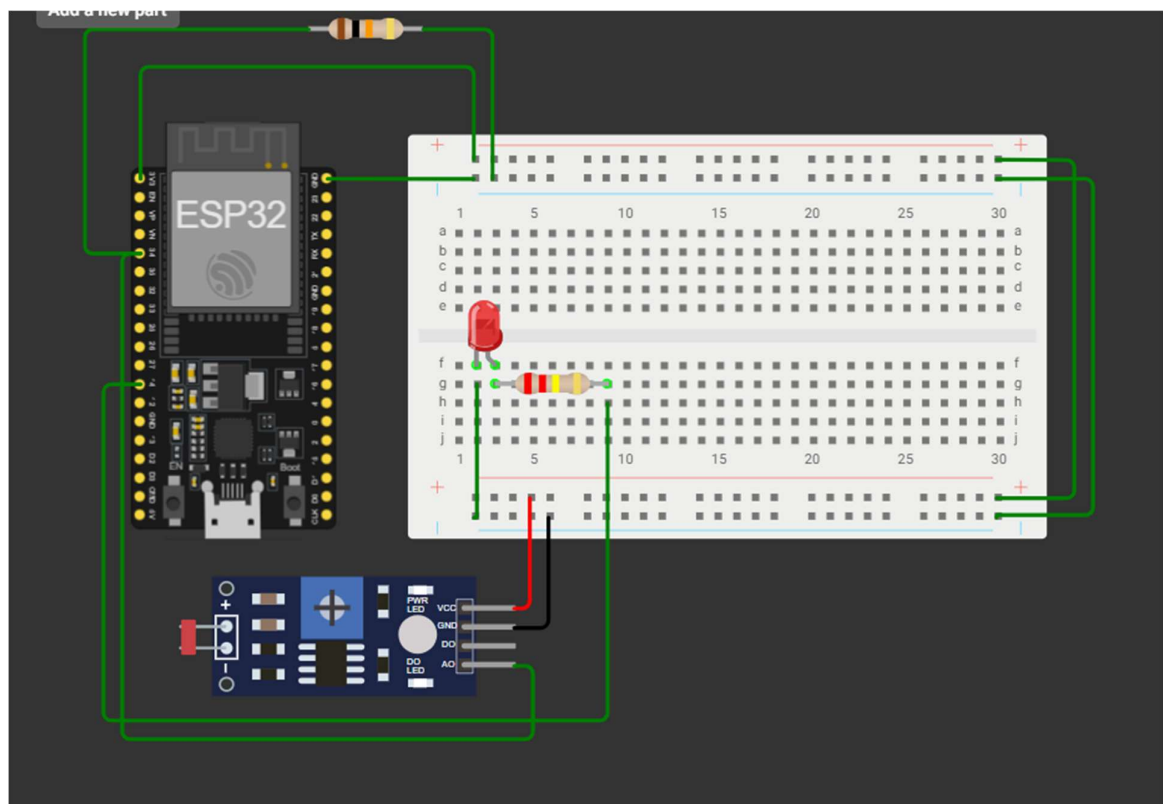
Q47. Heart Rate & SpO2 Monitor (Simulation)

Problem Statement

Using a MAX30102 pulse oximeter module (or simulate with an LDR as a proxy sensor), build a health monitoring prototype that measures pulse rate and blood oxygen saturation, displays values, and alerts on abnormal readings. This is a simulation for learning sensor interfacing, not a medical device.

GITHUB LINK : <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/QUES47>

IMAGE:



WOKWI LINK : <https://wokwi.com/projects/469007572949943297>

Q48. Design Challenge: IoT for Jammu Smart City J&K Government is planning a Smart City project for Jammu. As an IoT engineer, propose a system to monitor and manage 3 of the following (choose any 3): Street lighting, Traffic flow, Parking availability, Air quality, Water supply pressure, Waste bin fill levels.

For each: • Identify the sensor(s) needed • Identify the microcontroller / connectivity module

- Describe the data flow: Sensor → Protocol → Cloud → Dashboard
- Estimate approximate cost per node (in INR) Present your answer as a structured proposal (500–700 words) uploaded as /final_project/smart_city_proposal.md in GitHub.

GITHUB LINK : <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/Q48>

Q49. Security Vulnerability Analysis You are given a poorly written IoT code snippet that connects an ESP8266 to a public Wi-Fi network. The code hardcodes the SSID and password, uses unencrypted HTTP (not HTTPS), publishes sensor data to a public MQTT broker without authentication, and never validates incoming commands. Identify at least 5 security vulnerabilities, explain the risk of each, and provide corrected code for at least 2 vulnerabilities.

GITHUB LINK : https://github.com/Subham827-all/-iot-summer-school-2026/blob/main/ASSIGNMENT/ASSIGNMENT3/Q49_SecurityVulnerability/Q49_SecurityVulnerability.md

Q50. Reflection & Learning Portfolio Describe what you built and what you learned. How will you apply IoT skills in your field of study or future career? What would you build next — describe your own IoT project idea with sensors, connectivity, and application. (Minimum 400 words, personal and reflective)

GITHUB LINK <https://github.com/Subham827-all/-iot-summer-school-2026/tree/main/ASSIGNMENT/ASSIGNMENT3/Q50>